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is the force constant. Therefore applied force is $\vec{F}_a = +k\vec{x}$. The positive sign implies that the applied force is in the direction of displacement $\vec{x}$. The spring force is an example of variable force as it depends on the displacement $\vec{x}$. Let the spring be stretched to a small distance $d\vec{x}$. The work done by the applied force on the spring to stretch it by a displacement $\vec{x}$ is stored as elastic potential energy.

$$U = \int \vec{F}_a . d\vec{r} = \int_0^x \left|\vec{F}_a\right| \left|d\vec{r}\right| \cos\theta \qquad (4.24)$$
$$= \int_0^x F_a dx \cos\theta$$

The applied force $\vec{F}_a$ and the displacement $d\vec{r}$ (i.e., here dx) are in the same direction. As, the initial position is taken as the equilibrium position or mean position, x=0 is the lower limit of integration.

$$U = \int_0^x kx dx \qquad (4.24)$$

$$U = k\left[\frac{x^2}{2}\right]_0^x \qquad (4.24)$$

$$U = \frac{1}{2}kx^2 \qquad (4.24)$$

If the initial position is not zero, and if the mass is changed from position x_i to x_f, then the elastic potential energy is

$$U = \frac{1}{2}k\left(x_f^2 - x_i^2\right) \qquad (4.25)$$

From equations (4.24) and (4.25), we observe that the potential energy of the stretched spring depends on the force constant k and elongation or compression x.

Note

The potential energy stored in the spring does not depend on the mass that is attached to the spring.

Force-displacement graph for a spring

Since the restoring spring force and displacement are linearly related as $F = -kx$, and are opposite in direction, the graph between F and x is a straight line with dwelling only in the second and fourth quadrant as shown in Figure 4.10. The elastic potential energy can be easily calculated by drawing a F - x graph. The shaded area (triangle) is the work done by the spring force.

$$\text{Area} = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2} \times (x) \times (kx).$$
$$= \frac{1}{2}kx^2$$

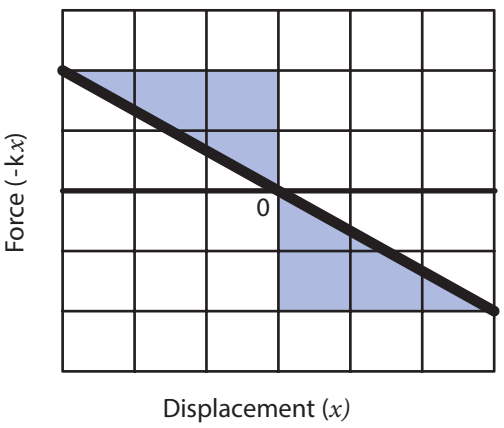


Figure 4.10 Force–displacement graph for a spring